PRAKTIKUM BASIS DATA

MODUL 8

Scripting SQL (Create Index, Create View, Analyze Table)

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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2026

1. **Design database untuk suatu Perusahaan**

   a. **Tabel Departmen**

Query :

```sql
CREATE TABLE departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(50) NOT NULL,
    manager_id INT,
    location_id INT
);
```

Hasil Query :

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID
1	DEPARTMENT_ID	NUMBER(38,0)	No	(null)	1
2	DEPARTMENT_NAME	VARCHAR2(50 BYTE)	No	(null)	2
3	MANAGER_ID	NUMBER(38,0)	Yes	(null)	3
4	LOCATION_ID	NUMBER(38,0)	Yes	(null)	4

b. Tabel Employees

Query :

```sql
CREATE TABLE employees (
    employee_id INT PRIMARY KEY,
    last_name VARCHAR(50) NOT NULL,
    email VARCHAR(100),
    salary DECIMAL(10, 2),
    commission_pct DECIMAL(5, 2),
    hire_date DATE NOT NULL,
    department_id INT,
    CONSTRAINT fk_department
        FOREIGN KEY (department_id)
        REFERENCES departments(department_id)
);
```

Hasil Query :

	COLUMN_NAME	DATA_TYPE	NULLABLE	DATA_DEFAULT	COLUMN_ID
1	EMPLOYEE_ID	NUMBER(38,0)	No	(null)	1
2	LAST_NAME	VARCHAR2(50 BYTE)	No	(null)	2
3	EMAIL	VARCHAR2(100 BYTE)	Yes	(null)	3
4	SALARY	NUMBER(10,2)	Yes	(null)	4
5	COMMISSION_PCT	NUMBER(5,2)	Yes	(null)	5
6	HIRE_DATE	DATE	No	(null)	6
7	DEPARTMENT_ID	NUMBER(38,0)	Yes	(null)	7

c. **Tabel Suppliers**

Query :

```sql
CREATE TABLE suppliers (
    supplier_id INT PRIMARY KEY,
    supplier_name VARCHAR(100) NOT NULL,
    contact_name VARCHAR(50),
    phone VARCHAR(20)
);
```

Hasil Query :

SUPPLIER...	SUPPLIER...	CONTACT...	PHONE

Tabel products

Query :

```sql
CREATE TABLE products (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    supplier_id INT,
    unit_price DECIMAL(10, 2),
    stock_quantity INT,
    CONSTRAINT fk_supplier
        FOREIGN KEY (supplier_id)
        REFERENCES suppliers(supplier_id)
);
```

Hasil Query :

PRODUCT...	PRODUCT...	SUPPLIER...	UNIT_PRICE	STOCK_Q...

2. Analisis Table Menggunakan Compute Statistic

Query :

```sql
--2
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
```

Hasil Query :

```
Table DEPARTMENTS analyzed.


Table EMPLOYEES analyzed.
```

3. Isi data untuk table

Table Departments

Query :

```
--3
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (10, 'Administration', 20
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (20, 'Marketing', 201, 18
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (30, 'Purchasing', 202, 1
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (40, 'Human Resources', 2
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (50, 'Shipping', 204, 150
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (60, 'IT', 205, 1400);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (70, 'Public Relations',
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (80, 'Sales', 207, 2500);
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (90, 'Executive', 208, 17
INSERT INTO departments (department_id, department_name, manager_id, location_id) VALUES (100, 'Finance', 209, 170
```

Hasil Query :

	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	Administration	200	1700
2	20	Marketing	201	1800
3	30	Purchasing	202	1700
4	40	Human Resources	203	2400
5	50	Shipping	204	1500
6	60	IT	205	1400
7	70	Public Relations	206	2700
8	80	Sales	207	2500
9	90	Executive	208	1700
10	100	Finance	209	1700

Tabel Employees

Query :

```
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (1, 'King', 'king@mail.com', 5000, 0.1, TO_DATE('2023-01-01', 'YYYY-MM-DD'), 90);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (2, 'Kochhar', 'kochhar@mail.com', 4500, NULL, TO_DATE('2023-02-15', 'YYYY-MM-DD'), 90);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (3, 'Abel', 'abel@mail.com', 3800, 0.2, TO_DATE('2023-03-10', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (4, 'Taylor', 'taylor@mail.com', 3200, 0.15, TO_DATE('2023-04-05', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (5, 'Grant', 'grant@mail.com', 2800, 0.1, TO_DATE('2023-05-20', 'YYYY-MM-DD'), 80);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (6, 'Hunold', 'hunold@mail.com', 4000, NULL, TO_DATE('2023-01-20', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (7, 'Ernst', 'ernst@mail.com', 3500, NULL, TO_DATE('2023-02-10', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (8, 'Lorentz', 'lorentz@mail.com', 3000, NULL, TO_DATE('2023-03-01', 'YYYY-MM-DD'), 60);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (9, 'Whalen', 'whalen@mail.com', 2500, NULL, TO_DATE('2023-04-12', 'YYYY-MM-DD'), 10);
INSERT INTO employees (employee_id, last_name, email, salary, commission_pct, hire_date, depart
VALUES (10, 'Zlotkey', 'zlotkey@mail.com', 3100, 0.1, TO_DATE('2023-05-01', 'YYYY-MM-DD'), 80);
```

Hasil Query :

	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPART
1	1	King	king@mail.com	5000	0,1	01-01-2023	
2	2	Kochhar	kochhar@mail.com	4500	(null)	15-02-2023	
3	3	Abel	abel@mail.com	3800	0,2	10-03-2023	
4	4	Taylor	taylor@mail.com	3200	0,15	05-04-2023	
5	5	Grant	grant@mail.com	2800	0,1	20-05-2023	
6	6	Hunold	hunold@mail.com	4000	(null)	20-01-2023	
7	7	Ernst	ernst@mail.com	3500	(null)	10-02-2023	
8	8	Lorentz	lorentz@mail.com	3000	(null)	01-03-2023	
9	9	Whalen	whalen@mail.com	2500	(null)	12-04-2023	
10	10	Zlotkey	zlotkey@mail.com	3100	0,1	01-05-2023	

4. Analisis Table data setelah di isi data

Query :

```
ANALYZE TABLE departments COMPUTE STATISTICS;
ANALYZE TABLE employees COMPUTE STATISTICS;
```

Hasil Query :

```
Table DEPARTMENTS analyzed.


Table EMPLOYEES analyzed.
```

5. View Empvu80

Query :

```
CREATE OR REPLACE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employees
WHERE department_id = 80;
```

Hasil Query :

```
View EMPVU80 created.
```

6. Tampilkan empvu80

Query :

```
DESC empvu80;
```

Hasil Query :

```
Name           Null?    Type
-------------- -------- ------------
ID_NUMBER      NOT NULL NUMBER(38)
NAME           NOT NULL VARCHAR2(50)
SALARY                  NUMBER(10,2)
DEPARTMENT_ID           NUMBER(38)
```

7. Non unique-index pada kolom FK pada table employee

Query :

```
CREATE INDEX emp_dept_idx
ON employees(department_id);
```

Hasil Query :

```
Index EMP_DEPT_IDX created.
```